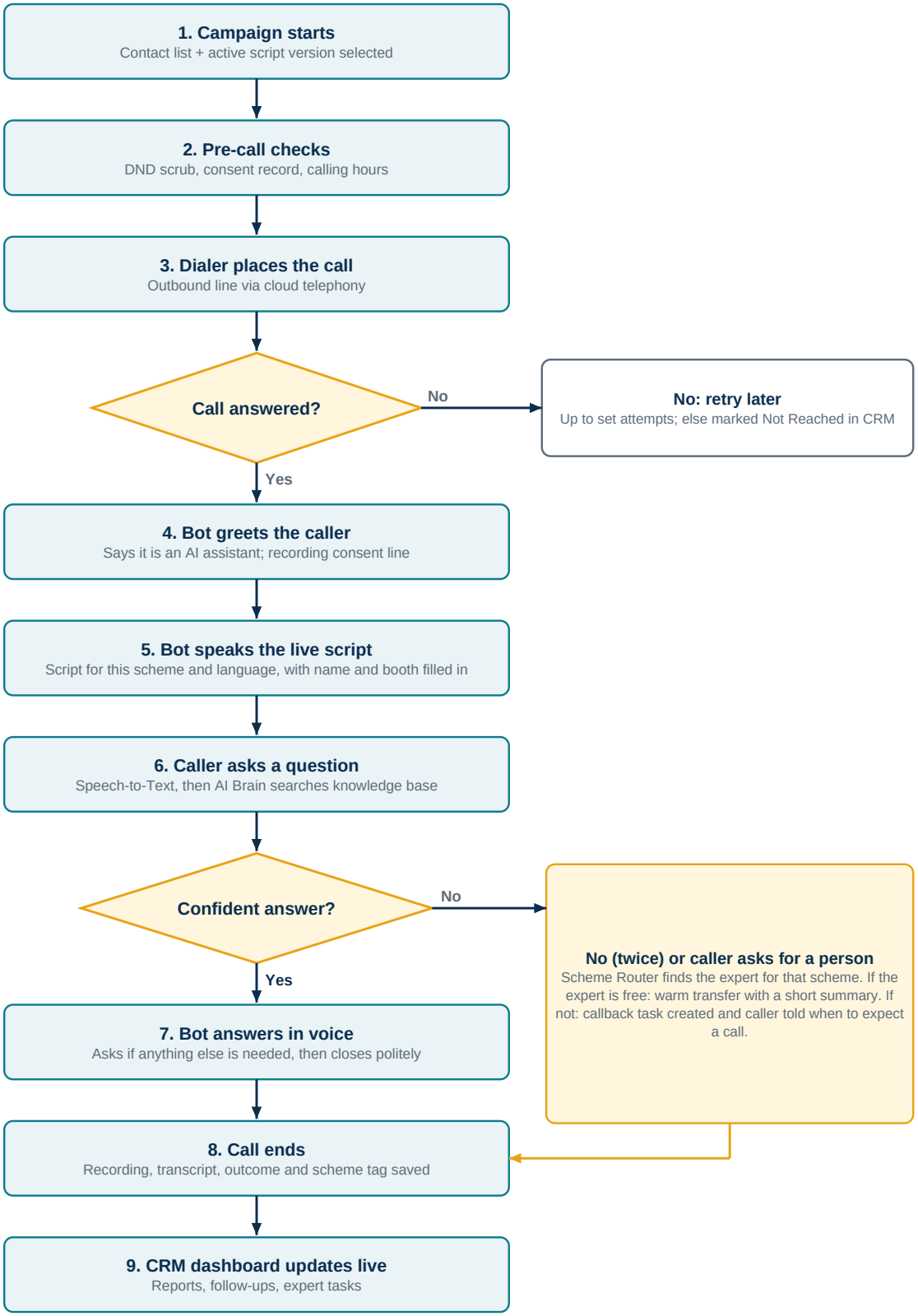
System Architecture

AI Voice Call Centre + WhatsApp + CRM Dashboard for BJP Call Center for Public Seva | Document MORYA-AICC-ARCH-02 | 19 October 2026

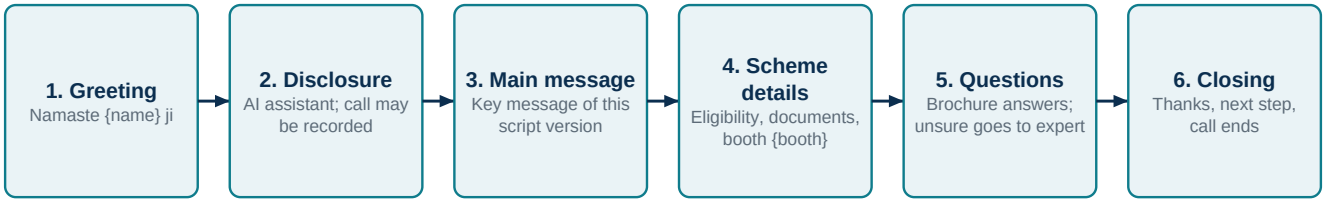
Every call passes through the same chain: telephony and compliance checks, the voice AI engine, the live script, the approved knowledge base, and then into the CRM. The bot speaks according to the script the admin has added, and uses brochures only for questions outside the script.

Call Flow: How One Call Works



Script-Driven Bot: How the Bot Speaks According to the Script

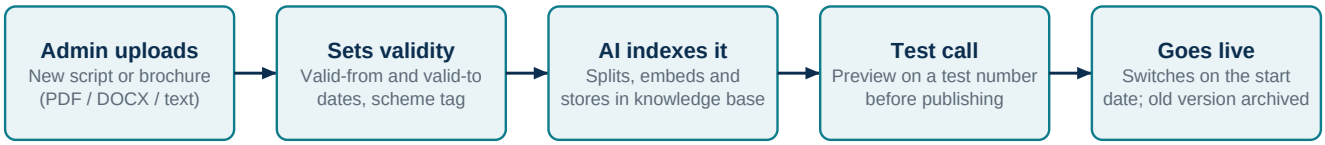
The admin adds the call script in the dashboard. The bot speaks that script line by line, in the right language, with each caller's details filled in. Brochures are only used when the caller asks something the script does not cover.



What the admin controls in the script editor	How it works
Script text per scheme	Separate script for each scheme (health, education, farmer, voter help)
Language	Hindi, Marathi and English versions of the same script
Caller details	Variables such as {name}, {booth}, {scheme} are filled in from the contact list
Voice and speed	Choose the voice, speaking speed and pauses
Valid from / to	A new script can be scheduled; it goes live on the start date and the old one is archived
Test and approve	Test call to a number before publishing; version history with rollback
Brochures	Uploaded PDFs become the knowledge for questions outside the script

Rule: script first, knowledge second. If the caller's question is not in the script or the brochures, the call goes to the scheme expert and the bot never guesses.

Script and Brochure Update Flow (every 15 to 20 days)



Scheme-wise Calling and Expert Hand-off

Scheme	Bot explains	Expert mapped for hand-off
Health support	Eligibility, hospitals, documents to carry	One named expert per scheme
Education / scholarship	Last dates, marksheet and income documents	One named expert per scheme
Farmer support	Benefits, forms, nearest centre	One named expert per scheme
Voter help / booth info	Booth location, documents to carry (Voter ID, Aadhaar, PAN)	One named expert per scheme

Technology Stack

Layer	Technology
AI brain	Claude LLM with retrieval from the approved knowledge base
Speech	Indian-language speech-to-text and text-to-speech (Hindi, Marathi, English)
Telephony	Exotel or Twilio, DLT-registered route with call recording
WhatsApp	Official WhatsApp Business API through an approved provider
Backend / database	Python (FastAPI) or Node.js, PostgreSQL, vector database
Dashboard	React web app with role-based login (Admin, Supervisor, Expert)
Hosting	India-region cloud with encrypted storage, daily backups

Security and Compliance Built In

- AI disclosure and recording consent line at the start of every call.
- DND scrub, calling-hour limits and consent records kept before dialing.
- Aadhaar is never read out or stored in full; only the last four digits are confirmed.
- Role-based access, encrypted recordings, audit log of every admin action.
- The bot answers only from approved scripts and brochures; anything else goes to the scheme expert.